

Building a Python Adventure Game with GitHub Copilot

Course-End Project Write-Up

Introduction

This project involves developing a text-based adventure game using Python. The goal is to strengthen core programming skills such as functions, loops, conditionals, and user input handling while creating an interactive command-line application. GitHub Copilot was used as an AI assistant to support code generation, improve efficiency, and speed up development. The game allows players to explore different locations, make decisions, and complete a quest to find a hidden treasure.

Project Objective

The primary objective of this project was to design a fun and interactive adventure game while applying fundamental Python concepts. The game required creating multiple decision paths where user choices directly affect the outcome. Another key objective was to learn how GitHub Copilot can assist in writing optimized and structured code.

Game Structure and Design

The game begins by introducing the storyline and asking the player to enter their name. The player then chooses between two paths: exploring a dark forest or entering a mysterious cave. Each path leads to different challenges and decisions.

Separate functions were created for each section of the game:

- `start_game()` to introduce the story and take user input
- `forest_path()` to handle forest-related events

- `cave_path()` to manage cave scenarios

Conditional statements determine the outcomes of player choices. The game can end in three ways:

1. Winning by finding the treasure
2. Losing due to incorrect decisions
3. Restarting the game to try again

A loop ensures the game continues until the player completes the adventure or exits.

Use of GitHub Copilot

GitHub Copilot played an important role during development. It helped by suggesting function templates, generating conditional logic, and recommending cleaner code structures. Copilot reduced development time by automating repetitive tasks and providing helpful code completions. It also helped improve readability and efficiency of the program.

Challenges Faced

During development, some challenges were encountered, such as managing multiple story branches, handling invalid user inputs, and ensuring smooth transitions between functions. These issues were resolved by adding proper input validation, clear conditional checks, and systematic testing of each function.

Enhancements

Additional improvements were added to enhance the gameplay experience. These include better prompts, clear instructions, and a replay option after the game ends. The code was organized into functions to maintain readability and modularity. Future

enhancements could include adding an inventory system, multiple levels, random events, or a graphical interface.

Learning Outcomes

This project helped reinforce important programming concepts such as modular design, conditional logic, loops, and user interaction. It also demonstrated how AI tools like GitHub Copilot can assist developers in writing faster and more efficient code. Overall, the project improved both problem-solving and coding skills.

Conclusion

The Python adventure game project successfully combines learning with creativity. It demonstrates how fundamental programming concepts can be applied to build an engaging application. With the assistance of GitHub Copilot, the development process became more efficient and productive. This project provided valuable hands-on experience in designing and implementing a complete Python program.